

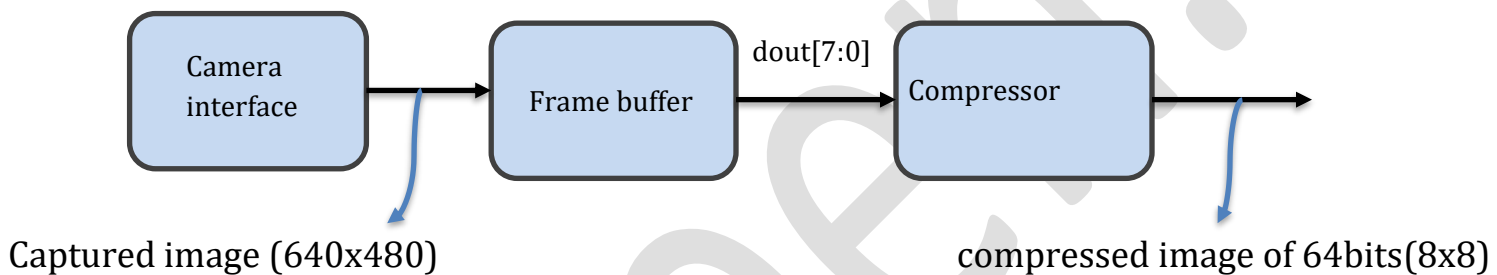
Camera Interfacing Project Documentation

Project Title:

640x480 to 8x8 Image Compression using OV7670 Camera and Verilog

Objective:

To capture a grayscale 640x480 image from an OV7670 camera and compress it into a binary 8x8 image using spatial downsampling and threshold-based binarization. The goal is to reduce the image to 64 bits while preserving basic structure.



Overview:

Input: OV7670 camera (640x480 grayscale pixels)

Output: 64-bit compressed binary image (8x8)

Processing Stages:

1. Pixel Capture using ov7670_capture module
2. Downsampling & Thresholding using image_compressor module

Understanding Pixels and Resolution

What does 640x480 mean?

- 640 = Number of columns (width or horizontal pixels)
- 480 = Number of rows (height or vertical pixels)
- So, the total number of pixels per frame = $640 \times 480 = 307,200$ pixels (standard VGA resolution)

What is a Pixel?

- A pixel is the smallest unit of a digital image.
- Each pixel holds a value representing a color or brightness level.
- In grayscale, each pixel uses 8 bits:

- 0 → black
- 255 → white
- In between → varying shades of gray (e.g., 128 is medium gray)

What is 8x8 in this Project?

We compress the 640x480 image into just 8 rows × 8 columns = 64 pixels

Each pixel in the 8x8 output image is represented by 1 bit (0 or 1), resulting in a 64-bit binary output.

Compression Technique Overview

1. Spatial Downsampling

What is it?

- Reducing resolution by sampling fewer pixels from a large image.

How?

- Divide the 640x480 image into 64 blocks (8×8 grid).
- Each block = 80×60 pixels → $640 / 8 = 80$, $480 / 8 = 60$
- From each 80×60 block, select one pixel (e.g., top-left or center)

Result: One representative pixel per block → total of 64 sampled pixels

Visualization:

Full Image (640x480)

[Block 1] [Block 2] ... [Block 8]

[Block 9] [Block10] ... [Block16]

...

[Block57].....[Block64]

↓ Pick 1 pixel from each block ↓

Compressed Image (8x8)

[P1] [P2] ... [P8]

[P9] [P10]... [P16]

...

[P57]..... [P64]

2. Threshold-Based Binarization

What is it?

- Converts each 8-bit grayscale pixel into a 1-bit binary value.

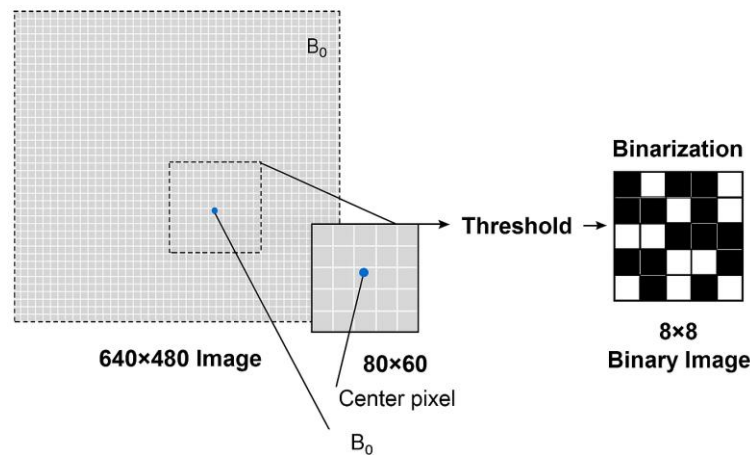
How?

- Set a threshold (e.g., 127)
- If pixel > threshold → 1 (white)
- Else → 0 (black)

Example:

Grayscale:		Binarized:
120 200 30		0 1 0
250 90 128	→	1 0 1
60 180 255		0 1 1

Final Compressed Output:



compressed_image = 64'b010100...011



Future Enhancements

- Use block average instead of single pixel
- Adaptive thresholding techniques (e.g., Otsu's method)
- Integrate external memory (BRAM/SDRAM) for image storage
- Add UART/SPI interface for data transfer

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